

A Reversible Smoothing Framework for Degenerate Riemannian Metrics

A Finite-Dimensional Analogue of Nash–Moser Theory via Lie Group Actions

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Abstract

In [1] we gave a link between the Nash Embedding Theorem and the barren plateaus problem of quantum computing. In this paper we give more evidence of this link. We develop a finite-dimensional analogue of the Nash–Moser smoothing paradigm for the study of degenerate Riemannian metrics. Given a smooth parameter manifold M mapped into a Riemannian ambient space $(\mathcal{N}, h)$ via an embedding function $f_0 : M \rightarrow \mathcal{N}$, the induced pullback metric $g = f_0^* h$ may be rank-deficient on large regions (analogous to the “loss of derivatives” or “barren plateaus”).

We construct a family of reversible smoothing operators by perturbing the embedding along the flow of an ambient Lie group action:

$$f_\varepsilon(x) = \Phi_\varepsilon(f_0(x)) \tag{10}$$

whose perturbed pullback metrics

$$g_\varepsilon = f_\varepsilon^* h \tag{9}$$

become strictly positive-definite for sufficiently small $\varepsilon > 0$.

This produces a reversible metric inflation: a temporary embedding of (M, g) into the higher-dimensional ambient space $\mathcal{N}$, geometrically expanding the degenerate locus into a positively curved auxiliary topology (analogous to a bouquet of spheres), restoring non-degeneracy while preserving reversibility.

The construction is inspired by a reversible deformation mechanism originally arising in a quantum-algorithmic context. Here we extract and formalize the underlying geometric structure using Lie algebras, proving new results on:

- reversible extrinsic smoothing flows
- auxiliary metric inflation via commutator bounds
- a finite-dimensional “reverse Nash flow”

The resulting framework establishes a rigorous, purely geometric dictionary for regularizing degenerate optimization landscapes.

Introduction

This work is inspired by a reversible deformation mechanism introduced in a quantum-algorithmic setting in [1]. A unitary perturbation was used to escape regions of vanishing curvature (“barren plateaus”) in a variational landscape. Here we extract the underlying geometric mechanism and develop a fully mathematical formulation independent of quantum mechanics.

We study smooth manifolds M mapped into an ambient space whose induced Riemannian metric g is degenerate on large subsets. Such degeneracies arise naturally in optimization, geometric statistics, and information geometry.

A fundamental restriction of differential geometry is that pulling back a degenerate metric by an intrinsic diffeomorphism on M cannot change its rank. To restore positive-definiteness, one must act extrinsically, inflating the manifold into a higher-dimensional ambient space. This mirrors John Nash’s use of ambient Euclidean spaces to resolve metric singularities.

Our goal: construct a reversible, extrinsic smoothing operator that temporarily inflates the degenerate metric into a strictly positive-definite form utilizing Lie group actions.

Main contributions:

- An extrinsic smoothing operator f_ε acting via an ambient Lie group.
- A reversible metric inflation theorem:

$$g_\varepsilon = f_\varepsilon^* h$$

becomes strictly positive-definite for small ε .

- A geometric interpretation of (M, g_ε) as a bouquet of hyperspheres.
- A reverse Nash flow resolving the “loss of derivatives” via algebraic tame estimates.

Preliminaries and Definitions

2.1 The Ambient Space and Lie Group Action

Let $(\mathcal{N}, h)$ be an N -dimensional Riemannian manifold (the ambient space). Let $\mathcal{G}$ be a Lie group acting isometrically on $\mathcal{N}$, and let $\mathfrak{g}$ be its Lie algebra. For any $A \in \mathfrak{g}$ and $p \in \mathcal{N}$, the infinitesimal action generates a fundamental vector field on $\mathcal{N}$, denoted by $A \cdot p$.

2.2 The Parameter Manifold and Target Map

Let M be a smooth manifold of dimension d . Let $\mathcal{U} : M \rightarrow \mathcal{G}$ be a smooth map associating each point in M to a group element.

Fix a base point $p_0 \in \mathcal{N}$.

The target embedding is the smooth map $f_0 : M \rightarrow \mathcal{N}$ defined by:

$$f_0(x) = \mathcal{U}(x) \cdot p_0 \quad (8)$$

2.3 The Degenerate Baseline Metric

The baseline metric on M is the pullback of the ambient metric:

$$g = f_0^* h \quad (1)$$

For $v \in T_x M$, the differential is $df_0(v) = d\mathcal{U}(v) \cdot p_0$.

Defining the local Lie algebra generator $A(v) = \mathcal{U}(x)^{-1} d\mathcal{U}(v) \in \mathfrak{g}$, the metric evaluates to:

$$g(v, v) = h(A(v) \cdot p_0, A(v) \cdot p_0) \quad (2)$$

The metric g is degenerate at x if there exists a non-zero $v \in T_x M$ such that $A(v) \cdot p_0 = 0$. The subset where this occurs is the degeneracy locus, $D \subseteq M$.

2.4 Tame Non-Commutativity Condition

To resolve the degeneracy, we select a global smoothing generator $V \in \mathfrak{g}$.

The Lie bracket in $\mathfrak{g}$ is $[V, A]$.

We say V satisfies the tame non-commutativity estimate if, for all v in the null space of the baseline metric ($A(v) \cdot p_0 = 0$), the commutator does not annihilate the base point:

$$\|[V, A(v)] \cdot p_0\|_h \geq c\|v\| \quad (3)$$

for some structural constant $c > 0$. This inequality guarantees the injection of curvature into the degenerate subspace.

Construction of the Extrinsic Smoothing Operator

We construct the smoothing operator not as a coordinate transformation on M , but as a continuous perturbation of the embedding into $\mathcal{N}$.

For $V \in \mathfrak{g}$ and intensity $\varepsilon > 0$, define the perturbed target map $f_\varepsilon : M \rightarrow \mathcal{N}$:

$$f_\varepsilon(x) = \mathcal{U}(x) \exp(\varepsilon V) \cdot p_0 \quad (4)$$

At $\varepsilon = 0$, $f_0(x)$ is recovered. The inverse flow $\varepsilon \rightarrow 0$ perfectly deflates the perturbation.

3.1 Pullback of the Inflated Metric

The perturbed metric on M is:

$$g_\varepsilon = f_\varepsilon^* h \quad (20)$$

For a tangent vector $v \in T_x M$, the perturbed differential is:

$$df_\varepsilon(v) = d\mathcal{U}(v) \exp(\varepsilon V) \cdot p_0 = \mathcal{U}(x) A(v) \exp(\varepsilon V) \cdot p_0 \quad (21)$$

Because $\mathcal{G}$ acts isometrically, the action of $\mathcal{U}(x)$ preserves the norm h . Thus, the quadratic form of the perturbed metric is:

$$g_\varepsilon(v, v) = h(A(v) \exp(\varepsilon V) \cdot p_0, A(v) \exp(\varepsilon V) \cdot p_0) \quad (22)$$

Auxiliary Metric Inflation

We now compute the metric expansion to prove inflation.

We rewrite the action on p_0 by inserting the identity $\exp(\varepsilon V) \exp(-\varepsilon V)$:

$$A(v) \exp(\varepsilon V) \cdot p_0 = \exp(\varepsilon V) (\exp(-\varepsilon V) A(v) \exp(\varepsilon V)) \cdot p_0 \quad (23)$$

Recognizing the Adjoint representation, $\text{Ad}_{\exp(-\varepsilon V)}(A(v)) = \exp(-\varepsilon V) A(v) \exp(\varepsilon V)$. Because $\exp(\varepsilon V)$ is an isometry, it can be dropped from the norm evaluation:

$$g_\varepsilon(v, v) = h(\text{Ad}_{\exp(-\varepsilon V)}(A(v)) \cdot p_0, \text{Ad}_{\exp(-\varepsilon V)}(A(v)) \cdot p_0) \quad (24)$$

4.1 The Taylor Expansion of the Adjoint

Expanding the Adjoint representation yields:

$$\text{Ad}_{\exp(-\varepsilon V)}(A(v)) = A(v) - \varepsilon[V, A(v)] + \mathcal{O}(\varepsilon^2) \quad (25)$$

Assume v is a degenerate direction, meaning $A(v) \cdot p_0 = 0$. The zero-order term vanishes entirely, isolating the commutator:

$$\text{Ad}_{\exp(-\varepsilon V)}(A(v)) \cdot p_0 = -\varepsilon[V, A(v)] \cdot p_0 + \mathcal{O}(\varepsilon^2) \quad (26)$$

Substituting this into the metric:

$$g_\varepsilon(v, v) = \varepsilon^2 \|[V, A(v)] \cdot p_0\|_h^2 + \mathcal{O}(\varepsilon^3) \quad (27)$$

By the tame condition, $\|[V, A(v)] \cdot p_0\|_h \geq c\|v\|$.

Thus:

$$g_\varepsilon(v, v) \geq c^2 \varepsilon^2 \|v\|^2 > 0 \quad (28)$$

The degenerate direction has been strictly inflated into a positively curved coordinate.

The Reverse Nash Flow

In classical NET, the iteration protocol introduces progressively high-frequency metric deformations to force an isometric embedding. Our framework executes an Inverse Nash Flow:

- $g_0 = g$ is pathologically degenerate.
- The flow f_ε acts as a non-local embedding functor, pushing the manifold out into the ambient Lie group symmetries.
- g_ε becomes strictly positive-definite, restoring differentiability and resolving the “loss of derivatives.”
- Linear reversion $\varepsilon \rightarrow 0$ systematically deflates the auxiliary scaffolding, crumbling the manifold back to its physical constraints.

Main Theorems

Theorem 6.1 ” Existence of the Positive-Definite Auxiliary Manifold

Let $g = f_0^* h$ be highly degenerate. If $V \in \mathfrak{g}$ satisfies the tame non-commutativity estimate $\|[V, A_i] \cdot p_0\|_h \geq c > 0$ for all local generators, then for a sufficiently small smoothing intensity $\varepsilon > 0$, the perturbed metric tensor g_ε of the auxiliary manifold is strictly positive-definite.

Theorem 6.2 ” Reversibility

The family of mappings f_ε satisfies $\lim_{\varepsilon \rightarrow 0} f_\varepsilon^* h = g$, ensuring that optimization over the auxiliary topology resolves deterministically back to the baseline constraints.

Theorem 6.3 ” Bouquet of Spheres Topology

For sufficiently small ε , the local geometry of (M, g_ε) near the degeneracy locus is diffeomorphic to a sub-manifold of $\mathcal{N}$ spanning the orthogonal commutator directions $[V, A_i] \cdot p_0$. Topologically, this acts as a bouquet of hyperspheres:

$$S^{(k_1)} \vee S^{(k_2)} \vee \dots \vee S^{(k_m)} \quad (29)$$

utilizing the higher-dimensional “ballooning” geometry purely as an optimization catalyst.

References

[1] Locate reference

Appendix A. Example: Resolving a Degenerate Metric on $SO(3)$

To visualize this geometrically, let $\mathcal{N} = \mathbb{R}^3$ with standard Euclidean metric h . Let $\mathcal{G} = SO(3)$. Let $p_0 = (0, 0, 1)^T$ (the North Pole).

Let $M = \mathbb{R}^2$ with coordinates (x, y) .

Define $\mathcal{U}(x, y) = \exp(xJ_z)\exp(yJ_z)$, where J_z generates rotations around the Z-axis.

The target map is:

$$f_0(x, y) = \exp((x + y)J_z) \cdot (0, 0, 1)^T = (0, 0, 1)^T \quad (\text{A1})$$

Because J_z stabilizes p_0 , the entire manifold M maps to a single degenerate point. The baseline metric is $g = 0$.

We choose the smoothing operator $V = J_x$ (rotation around X). The perturbed map is:

$$f_\varepsilon(x, y) = \exp((x + y)J_z) \exp(\varepsilon J_x) \cdot p_0 \quad (\text{A2})$$

The term $\exp(\varepsilon J_x) \cdot p_0$ tilts the pole away from the Z-axis by angle ε , creating a circle of radius $\sin(\varepsilon)$. The parameters (x, y) now actively trace this non-degenerate circle.

The metric expands exactly as predicted:

$$g_\varepsilon = \varepsilon^2(dx + dy)^2 + \mathcal{O}(\varepsilon^3) \quad (\text{A3})$$

The degenerate point has inflated into a well-conditioned geometric scaffold.

Appendix B. Theorem B1 (Global reachability via Lie-orbit branching)

Let (N, h) be a smooth Riemannian manifold, G a Lie group acting isometrically on N , and M a smooth parameter manifold with a smooth embedding

$$f_0 : M \rightarrow N, \quad g := f_0^*h \quad (\text{B1})$$

whose pullback metric g is allowed to be degenerate on a closed subset $\Sigma \subset M$ (the degeneracy locus). Let $V \in \mathfrak{g}$ be a fixed Lie algebra element and, for $\varepsilon \in \mathbb{R}$ small, define the perturbed embeddings

$$f_\varepsilon(x) := \Phi_\varepsilon(f_0(x)), \quad \Phi_\varepsilon := \exp(\varepsilon V) \in G, \quad (\text{B2})$$

with auxiliary metrics $g_\varepsilon := f_\varepsilon^*h$.

Let $F : M \rightarrow \mathbb{R}$ be a smooth cost function. Consider the piecewise-smooth dynamics obtained by alternating:

1. gradient flow of F with respect to g_ε on each auxiliary branch, and
2. reversible “inflation–deflation” steps $\varepsilon \mapsto 0$ that return the state to Σ when a plateau or non-global local minimum is detected.

Assumptions

(A1) Commutator tame estimate on the degeneracy locus: For every $x \in \Sigma$ and every nonzero $v \in T_x M$ with $g(v, v) = 0$, the associated generator $A(v) \in \mathfrak{g}$ satisfies

$$\|[V, A(v)] \cdot p_0\|_h \geq c\|v\| \quad (\text{B3})$$

for some constant $c > 0$ and fixed base point $p_0 \in N$. Equivalently, the auxiliary metric g_ε is strictly positive-definite on Σ for all sufficiently small $\varepsilon \neq 0$.

(A2) Orbit transversality at the degeneracy locus: For every $x \in \Sigma$, the derivatives of the family $\{f_\varepsilon\}$ span the ambient tangent space along the image of x :

$$\text{span}\{df_\varepsilon(x)[T_x M] \mid \varepsilon \in (-\varepsilon_0, \varepsilon_0)\} = T_{f_0(x)} N. \quad (\text{B4})$$

In particular, the Lie orbit through $f_0(x)$ is transversal to the degenerate directions of g .

(A3) Bouquet coverage of the search space: There exists a finite or countable family of auxiliary “balloons” $\{B_k\}_{k \in K} \subset M$ such that:

- each B_k is an open subset on which some g_{ε_k} is strictly positive-definite and the gradient flow of F is well-posed;
- each B_k intersects Σ (i.e., $\Sigma \cap B_k \neq \emptyset$);
- the union $\bigcup_{k \in K} B_k$ covers M (or at least the sublevel set $\{F \leq F(x_0)\}$ of interest).

(A4) Unique global attractor: There exists a unique point $x^* \in M$ such that:

- x^* is the unique global minimizer of F ;
- for every auxiliary metric g_{ε_k} and every initial condition in the corresponding B_k , the negative gradient flow of F with respect to g_{ε_k} has ω -limit set contained in the basin of attraction of x^* (with respect to the original metric structure).

Then, for any initial point $x_0 \in M$, the piecewise-smooth “inflate–flow–deflate” dynamics described above is globally reachable in the following sense:

- every maximal trajectory has ω -limit set $\{x^*\}$;
- in particular, the global minimizer x^* is reached asymptotically as $t \rightarrow +\infty$.

Proof

Step 1: Local metric inflation near Σ . By assumption (A1) and your existing metric-inflation theorem (Theorem 6.1 in 5.pdf), for each $x \in \Sigma$ there exists $\varepsilon_0 > 0$ such that, for all $|\varepsilon| < \varepsilon_0$, the auxiliary metric g_ε is strictly positive-definite in a neighborhood of x . Thus, every degenerate direction of g at Σ is “inflated” into a non-degenerate direction for g_ε .

Step 2: Star-graph (bouquet) structure. By (A3), the family $\{B_k\}$ forms a collection of positively curved auxiliary regions, each intersecting Σ . Topologically, this realizes a “star-graph” or bouquet structure: Σ acts as a hub, and each B_k is a branch (or “balloon”) attached to it. The reversible inflation–deflation mechanism ensures that whenever a trajectory reaches Σ , it can be lifted into any of the intersecting B_k by choosing an appropriate ε_k .

Step 3: Escape from nonglobal critical points. Let x_{loc} be a strict local minimum of F that is not global. If the trajectory enters a neighborhood of x_{loc} inside some B_i , the gradient flow with respect to g_{ε_i} may converge to x_{loc} . By construction of the algorithm, once the “Iteration–Time–Stuck” (ITS) criterion detects stagnation (vanishing gradient or plateau), the reversible inflation–deflation step is triggered: the state is transported back along the Lie orbit to a point in Σ . Because the flow is reversible and norm-preserving at the level of the ambient action, this return does not erase the global structure of the landscape, only the local trapping.

Step 4: Transversal branching from Σ . By (A2), at each $x \in \Sigma$ the derivatives of the embeddings f_ε span the ambient tangent space. In particular, the Lie orbit through $f_0(x)$ provides a transversal cone of directions that are not tangent to the degenerate subspace. Combined with (A1), this implies that for each visit to Σ there exists at least one auxiliary branch B_k such that the lifted point lies in a region where the gradient of F (with respect to g_{ε_k}) is nonzero and points away from previously explored basins of attraction. Thus, the algorithm can “branch” into a new balloon B_k that is dynamically distinct from the previously visited one.

Step 5: Coverage and eventual entry into the basin of x^* . By (A3), the union of balloons $\bigcup_k B_k$ covers the relevant portion of the search space. By (A4), the global minimizer x^* is the unique global attractor, and its basin of attraction is nonempty and invariant under the gradient flows induced by the auxiliary metrics. Therefore, there exists at least one index k^* such that B_{k^*} intersects the basin of x^* . Because each time the trajectory is trapped in a nonglobal basin it is returned to Σ and then branched into a (possibly new) balloon, and because the balloons cover the space and the Lie orbit is transversal, the process cannot be confined forever to basins that do not contain x^* . Hence, after finitely many inflation–deflation cycles, the trajectory enters B_{k^*} in a point lying in the basin of attraction of x^* .

Step 6: Asymptotic convergence. Once the trajectory lies in the basin of x^* for some auxiliary metric $g_{\varepsilon_{k^*}}$, the negative gradient flow of F with respect to $g_{\varepsilon_{k^*}}$ converges to x^* as $t \rightarrow +\infty$ by (A4). The reversibility of the smoothing flow ensures that subsequent deflation steps do not destroy this convergence:

they only change the metric representation, not the location of the minimizer. Therefore, the ω -limit set of the full piecewise-smooth dynamics is $\{x^*\}$.

This proves global reachability of the unique global minimizer under assumptions (A1)–(A4). $\square$